

Module 4.5: Statistical Reasoning - Key Terms & Definitions, PART 5

Alternative Hypothesis: The alternative hypothesis usually represents what we want to check, or what we suspect is really going on.

Confidence Interval: A range computed using sample statistics to estimate an unknown population parameter with a stated level of confidence.

Understanding the general structure of the confidence intervals for different levels of confidence and found that, in general, it has the form

where z^* is a general notation for the multiplier that depends on the level of confidence

For a 90% level of confidence, $z^* = 1.645$

For a 95% level of confidence, $z^* = 2$ (or 1.96 if you want to be precise)

For a 99% level of confidence, $z^* = 2.576$

To start our discussion about the structure of the confidence interval, let's consider the case of a population mean

The confidence interval, then, has the form: $\bar{x} \pm m$



$\bar{x}$ is the sample mean, the point estimator for the

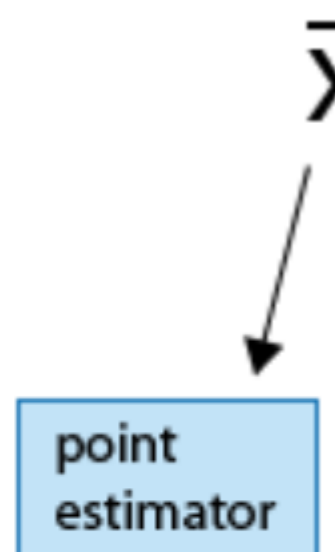
m is called the **margin of error**, since it represents

For example, for a 95% confidence interval, we mean by more than m , the margin of error.

m is further made up of the product of two components

- z^* , the **confidence multiplier**, and
- $\frac{\sigma}{\sqrt{n}}$, which is the standard deviation of $\bar{X}$,

Here is a summary of the different components



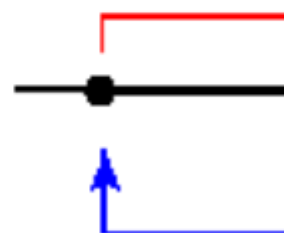
This structure:

esti

where the margin of error is further composed of (as we'll see, the standard error) is the general structure

Obviously, even though each confidence interval is different from confidence interval to confidence interval, it all aims to estimate.

Since the structure of the confidence interval is symmetric and centered at the estimate (in our case, $\bar{x}$), and its width



The margin of error, m , is therefore “in charge” of its width. The estimate, $\bar{x}$, is in charge of its location (and has no effect on the width).

- Example I:

The statistics professors at a university want to estimate the average statistics anxiety score for all of their undergraduate students. It would be too time consuming and costly to give every undergraduate student at the university their statistics anxiety survey. Instead, they take a random sample of 50 undergraduate students at the university and administer their survey.

Using the data collected from the sample, they construct a 95% confidence interval for the mean statistics anxiety score in the population of all university undergraduate students. They are using $\bar{x}$ to estimate μ . If the 95% confidence interval for μ is 26 to 32, then we could say, "we are 95% confident that the mean statistics anxiety score of all undergraduate students at this university is between 26 and 32." In other words, we are 95% confident that $26 \leq \mu \leq 32$. This may also be written as $[26, 32]$.

- Example II:

-

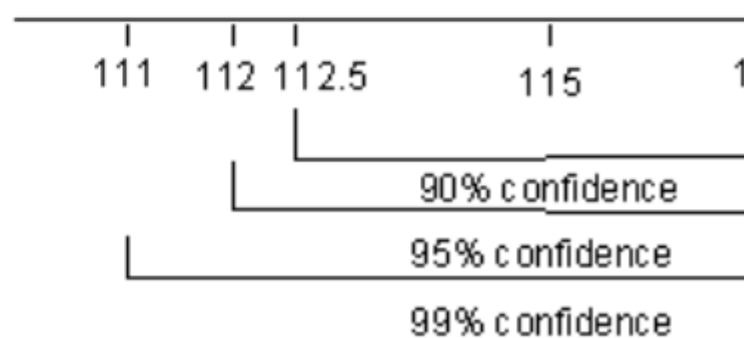
The IQ level of students at a particular university has an unknown μ and $\sigma = 15$. A simple random sample of 100 students is found to have $\bar{x} = 115$. Compute 90%, 95%, and 99% confidence intervals.

A 90% confidence interval for μ is $\bar{x} \pm 1.645 \frac{\sigma}{\sqrt{n}} = 115 \pm 1.645 \left(\frac{15}{\sqrt{100}} \right)$.

A 95% confidence interval for μ is $\bar{x} \pm 2 \frac{\sigma}{\sqrt{n}} = 115 \pm 2 \left(\frac{15}{\sqrt{100}} \right)$.

A 99% confidence interval for μ is $\bar{x} \pm 2.576 \frac{\sigma}{\sqrt{n}} = 115 \pm 2.576 \left(\frac{15}{\sqrt{100}} \right)$.

Note that the more confidence I require, the wider the confidence interval is (the “margin of error”). The 99% confidence interval is wider than the 95% confidence interval.



Confidence Intervals for μ

Confidence Intervals for the Population Mean:

- When the population is normal and/or the sample is large, a confidence interval for unknown μ when σ is known is:

$$\bar{x} \pm z^* \frac{\sigma}{\sqrt{n}},$$

where z^* is 1.645 for 90% confidence, 2 for 95% confidence, and 2.576 for 99% confidence.

- There is a trade-off between the level of confidence and the precision of the interval estimation. To pay for more precision is sacrificing level of confidence.
- The general form of confidence intervals is an estimate $\pm$ the margin of error (m). In this case, $m = z^* \frac{\sigma}{\sqrt{n}}$. The confidence interval is therefore centered at the estimate and its width is exactly $2m$.
- For a given level of confidence, the width of the interval depends on the sample size. We can therefore do a sample size calculation to figure out what sample size is needed in order to get a confidence interval with a certain margin of error m , and a certain level of confidence (assuming we have some flexibility with the sample size calculation we use:

$$n = \left(\frac{z^* \sigma}{m} \right)^2$$

(and round *up* to the next integer).

- [Optional Reading] When σ is unknown, we use the sample standard deviation, s , instead, but as we will see, we need to use a different set of confidence multipliers (t^*) associated with the t distribution. The interval is then:

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

- [Optional Reading] These new multipliers have the added complexity that they depend not only on the level of confidence, but also on the sample size. Software is therefore very useful for calculating confidence intervals in this case.
- [Optional Reading] For large values of n , the t^* multipliers are not that different from the z^* multipliers, so we can therefore use the interval formula:

$$\bar{x} \pm z^* \frac{s}{\sqrt{n}}$$

for μ when σ is unknown provides a pretty good approximation.

Hypothesis Testing: We say that hypothesis testing is “the other kind of inference”

because, unlike the inferential methods we presented so far, where the goal was estimating the unknown parameter, the idea, logic and goal of hypothesis testing are quite different.

- In **hypothesis testing**, we have some claim about the population, and we check whether or not the data obtained from the sample provide evidence against this claim

Example (C) 1

It was claimed that among all U.S. adults, about half are in favor of stricter gun control and about 60% are against it. In a recent poll of a random sample of 1,200 U.S. adults, 60% were in favor of stricter gun control. This data, therefore, provides some evidence against the claim.

Example (D) 2

It is claimed that among drivers 18-23 years of age (our population) there is no relationship between drunk driving and gender. A roadside survey collected data from a random sample of 5,000 drivers and recorded whether they were drunk and whether they were male or female. The collected data showed roughly the same percent of drunk drivers among males and females.

and among females. These data, therefore, do not give us any reason to reject the claim that drunk driving is not related to gender.

More Details and Terminology Now that we understand the general idea of how statistical hypotheses testing works, let's go back to each of the steps and delve slightly deeper, getting more details and learning some terminology.

Hypothesis testing step 1: Stating the claims.

In all three examples, our aim is to decide between two opposing points of view, Claim 1 and Claim 2. In hypothesis testing, *Claim 1* is called the *null hypothesis* (denoted " H_0 "), and *Claim 2* plays the role of the *alternative hypothesis* (denoted " H_a "). As we saw in the three examples, the null hypothesis suggests nothing special is going on, in other words, there is no change from the status quo, no difference from the traditional state of affairs, no contrast, the alternative hypothesis disagrees with this, stating that something is going on, or there is a change from the status quo, or there is a difference from the traditional state of affairs. The alternative hypothesis, H_a , is what we want to check or what we suspect is really going on.

Let's go back to our three examples and apply the new notation:

In example 1:

- H_0 : The proportion of smokers at Goodheart is 0.20.
- H_a : The proportion of smokers at Goodheart is less than 0.20.

In example 2:

- H_0 : The mean concentration in the shipment is the required 245 ppm.
- H_a : The mean concentration in the shipment is not the required 245 ppm.

In example 3:

- H_0 : Performance on the SAT is not related to gender (males and females score the same).
- H_a : Performance on the SAT is related to gender – males score higher.

Hypothesis testing step 2: Choosing a sample and collecting data.

This step is pretty obvious. This is what inference is all about. You look at sampled data in order to draw conclusions about the entire population. In the case of hypothesis testing, based on the data, you draw conclusions about whether or not there is enough evidence to reject H_0 .

There is, however, one detail that we would like to add here. In this step we collect data and *summarize* it. We will look at the second step in our three examples. Note that in order to summarize the data we used simple statistics such as the sample proportion $\hat{p}$, sample mean ($\bar{x}$) and the sample standard deviation (s).

In practice, you go a step further and use these sample statistics to summarize the data with what's called a test statistic. We are not going to go into any details right now, but we will discuss test statistics when we go through step 3.

Hypothesis testing step 3: Assessing the evidence.

As we saw, this is the step where we calculate how likely it is to get data like that observed when H_0 is true. This is the heart of the process, since we draw our conclusions based on this probability. If this probability is very small (see example 2), then that means that it would be very surprising to get data like that observed if H_0 were true. If we *did* observe such data is therefore evidence against H_0 , and we should reject it. On the other hand, if the probability is not very small (see example 3) this means that observing data like that observed is not very surprising, so the fact that we observed such data does not provide evidence against H_0 . This crucial probability, which has a special name. It is called the p -value of the test.

In our three examples, the p -values were given to you (and you were reassured that you didn't need to calculate these were derived):

- Example 1: p -value = 0.106
- Example 2: p -value = 0.0007
- Example 3: p -value = 0.29

Obviously, the smaller the p -value, the more surprising it is to get data like ours when H_0 is true, and the stronger the evidence the data provide against H_0 . Looking at the three p -values of our three examples, the data that we observed in example 2 provide the strongest evidence against the null hypothesis, followed by example 1, while the data in example 3 provides the least evidence against H_0 .

Inference For One Variable:

As we mentioned at the end of the introduction, the first part of the inference unit will deal with inference. Recall that in the Exploratory Data Analysis (EDA) unit, when we learned about summarizing the data of a variable (in the Examining Distributions module) we distinguished between two cases; categorical data and numerical data.

We will make a similar distinction here in the inference unit. In the EDA unit, the type of variable determined the statistical measures we used to summarize the data. In Inference, the type of variable of interest (categorical or quantitative) will determine what population parameter we are going to do inference for.

- When the variable of interest is *categorical*, the population parameter that we will infer about is the *proportion* (p) associated with that variable. For example, if we are interested in studying opinion on the death penalty among U.S. adults, and thus our variable of interest is “death penalty (in favor/against)”, we’ll choose a sample of U.S. adults and use the collected data to make an inference about p – the proportion of U.S. adults who support the death penalty.
- When the variable of interest is *quantitative*, the population parameter that we infer about is the *mean* (μ) associated with that variable. For example, if we are interested in studying the annual salaries of teachers in a certain state, we’ll choose a sample from that population and use the collected data to make an inference about μ , the mean annual salary of all teachers in that state.

Interval Estimation: In **interval estimation**, we estimate an unknown parameter using an interval of values that is likely to contain the true value of that parameter (and state how confident we are that this interval indeed captures the true value of the parameter).

- Example: Based on sample results, we are 95% confident that p , the proportion of U.S. adults who are in favor of stricter gun control, is between 0.57 and 0.63.

Null Hypothesis: The null hypothesis claims that “nothing special is going on” or “there is no change from the status quo.”

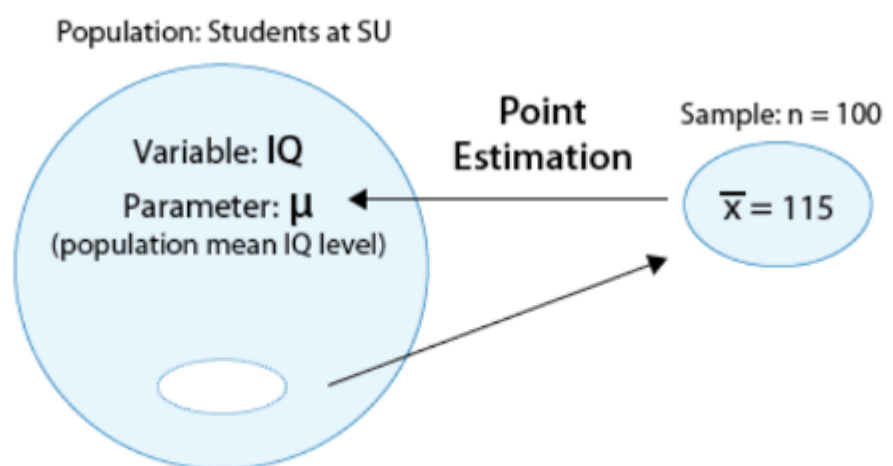
Point Estimation: Point estimation is the form of statistical inference in which, based on the sample data, we estimate the unknown parameter of interest using a single value (hence the name point estimation). We estimate an unknown parameter using a single

number that is calculated from the sample data.

- Example: Based on sample results, we estimate that p , the proportion of all U.S. adults who are in favor of stricter gun control, is 0.6.
- Example:

Suppose that we are interested in studying the IQ levels of students at Smart University (SU). (IQ level is a quantitative variable), we are interested in estimating μ , the mean IQ level of all students at SU. A random sample of 100 SU students was chosen, and their (sample) mean IQ level was found to be $\bar{x} = 115$. If we wanted to estimate μ , the population mean IQ level, by a single number based on the sample data,

it is intuitive to use the corresponding quantity in the sample, the sample mean $\bar{x} = 115$. We use $\bar{x}$ as the *point estimate* for μ , and in general, we'll always use $\bar{x}$ as the *point estimator* for μ . (Note that about the *specific value* (115), we use the term *estimate*, and when we talk in general about the term *estimator*). The following figure summarizes this example:



- **NOTE:** The larger the sample the point estimate is based on, the closer it is likely to be to the parameter it estimates.

Type I and Type II Errors:

What Can Go Wrong: Two Types of Errors Statistical investigations involve making decisions in the face of uncertainty. So there is always some chance of making a wrong decision. In hypothesis testing, the following decisions

- If the null hypothesis is true and we do not reject it, it is a *correct decision*.
- If the null hypothesis is false and we reject it, it is a *correct decision*.
- If the null hypothesis is true, but we reject it. This is a *type I* error.
- If the null hypothesis is false, but we fail to reject it. This is a *type II* error.

Type I and type II errors are not caused by mistakes. They are the result of random chance. The *data* point to a conclusion that is false. It's no one's fault!

NOTE: A type I error is possible only if we reject the H_0 . A type II error is possible only if we fail to reject the H_0 .